

Sablefish Assessment General Methodology

2026 CIE Review

Daniel Goethel and Matt Cheng

June 2026



Agenda

Tuesday, June 16, 2026

- A. Welcome and Introductions (30min)
- B. Sablefish Background (Management, Biology, and Data Inputs; 3hrs)
- C. General Assessment Model Structure (ToR 1; 3hrs)

Wednesday, June 17, 2026

- A. Reconvene and Review (30min)
- B. Summary of Closed Loop Simulations (3hrs)
- C. Comparing Assessment Performance and Diagnostics (ToRs 1/2; 3hrs)

Thursday, June 18, 2026

- A. Reconvene and Review (30min)
- B. Comparing Assessment Performance and Diagnostics *Cont'd* (ToR 1/2; 3hrs)
- C. Spatial Model Configurations and Diagnostics (ToR 3; 3hrs)

ADJOURN



Order of Operations

- **Now:** Background on general assessment framework common to all model types (**2026 SAFE, Appendix 3A**)
 - Age structured SCAA
 - SPoRC assessment platform
 - Key assumptions/structure
- **Tomorrow Morning:** Summary of closed loop simulations used to identify robust model structures (**Appendix 3E**)
- **Tomorrow Afternoon:** Model development (**Appendices 3B, 3C**) and final model results (**2026 SAFE, Appendix 3C**)
- **Wednesday:** Spatial model development, results, and ongoing work (**Appendix 3D**)

Documents For Review (available [here](#))

- **2026 Assessment of the Sablefish Stock in Alaska: 2026 Center for Independent Experts (CIE) Review Document ([here](#))** **Days 1-2**
- Appendix 3A. The Spatial Population over Regional Components (SPoRC) Model: Technical Description of the Assessment Model for Sablefish ([here](#))
- Appendix 3B. Model Building and Bridging Exercise for the Sablefish Single Region Models (Bridging from Model 25.12 to 26.1_1Reg) ([here](#)) **Day 2**
- Appendix 3C. Development of a Fleets-as-Areas (FAA) Stock Assessment Model for Alaska Sablefish (*Model 26.2 FAA*) ([here](#))
- **Appendix 3D. Results of Spatial Stock Assessment Models for Alaska Sablefish ([here](#))** **Day 3**
- **Appendix 3E. Evaluating Alternative Assessment Structures for Alaska Sablefish Under Survey Design Changes via Closed-Loop Simulation ([here](#))** **Day 2**



Assessment Biological and Fishery Considerations

- Long-lived (low M) and difficult to age (aging error matrix)
- Widely distributed (AK-wide assessment) but no population structure (single population)
- High mobility with ontogenetic migrations (differential availability to fishery/surveys across regions)
- Spasmodic and cyclic recruitment (lack of a well-defined SRR)
- Sexually dimorphic (sex-specific growth/selectivity)
- Multiple fisheries with different interactions with sablefish (multiple fleets with different selectivity)



Assessment Data and Modeling Considerations

- How best to handle recent changes in NOAA Longline Survey (LLS) design?
- Should a fleets-as-areas (FAA) model be used to avoid extrapolating LLS indices? (ToR 1+2)
- Are best practices for modeling sex- and fleet-structure being used in the assessment?
- Which data should be included in the assessment (e.g., indices, length compositions)?



The SPoRC Assessment Package

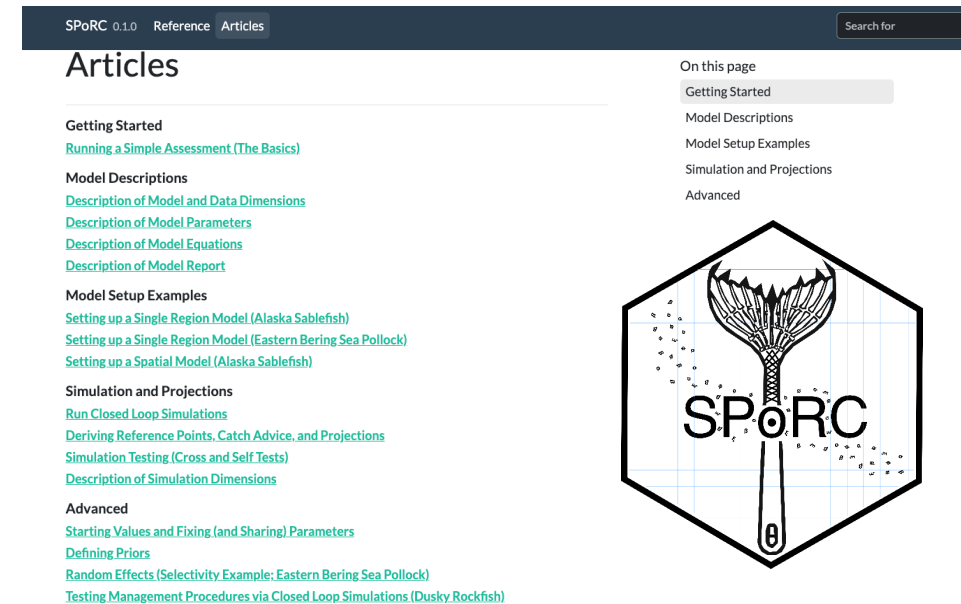
- Stochastic (S) Population (P) over (o) Regional (R) Components (C)
- Stock assessment platform written in R and R-TMB
- Generalized statistical catch-at-age model with spatial, sex, and fleet-structure
 - Supports random effects (Laplace Approximation) and Bayesian inference (rstan)
- Includes comprehensive stock assessment model diagnostics and simulation tools
- Thoroughly documented!



ORIGINAL ARTICLE **OPEN ACCESS**

The SPoRC Stock Assessment Package: A Generalized Next-Generation Platform to Assess Spatial, Age and Sex-Structured Populations

Matthew L.H. Cheng^{1,2} | Daniel R. Goethel¹ | Curry J. Cunningham² | Peter-John F. Hulson¹ | James N. Ianelli³ | Kristen L. Omori¹



<https://chengmatt.github.io/SPoRC/>



The SPoRC Assessment Package



- Motivated by the needs of Alaska sablefish
- Previous models were implemented using custom ADMB code
 - Difficult to update yearly and keep pace with research and fishery dynamics
- Needed a next-generation model that:
 - Incorporated sex-structure, fleet-structure, and spatial dynamics
 - Readily transitioned between single-region and multi-region applications
 - Efficiently estimated random effects
 - Was modular, flexible, and ‘easy’ to contribute to
 - Had a fully functional assessment wrapper – catch projections, diagnostics, residual analysis, plotting functions

TABLE 4 | Primary features of SPoRC in comparison to capabilities of existing assessment platforms.

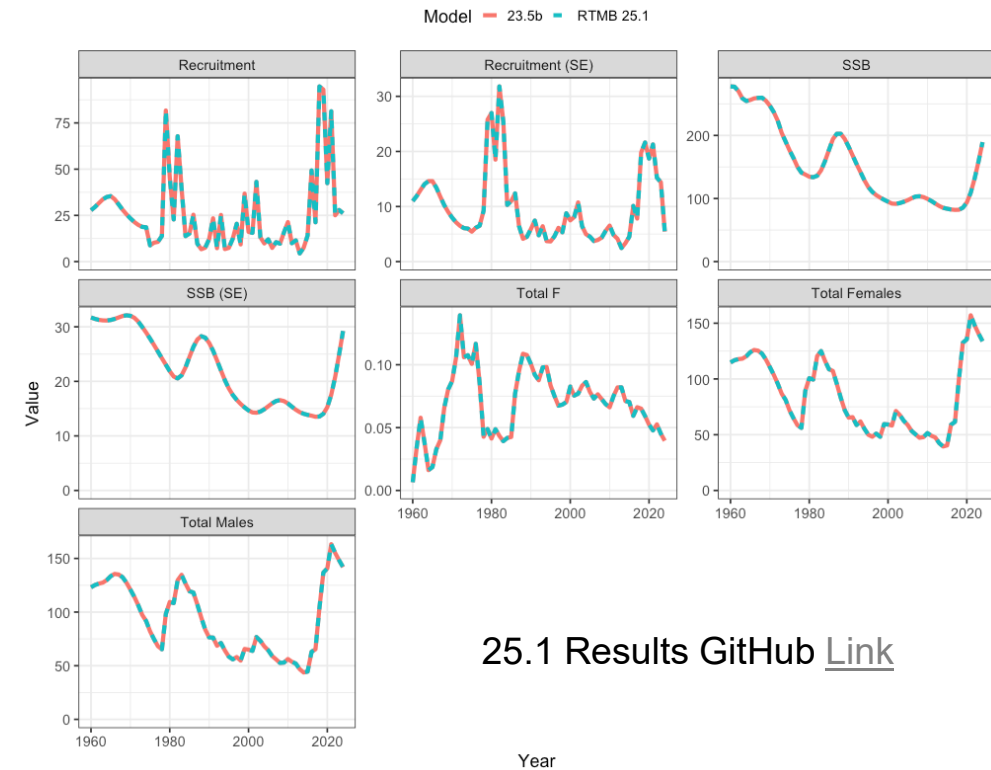
Process	Feature	Platforms with similar capabilities
Population structure	Panmictic, Spatially Heterogenous, and Metapopulation Structure	CASAL2, GADGET
Spatial structure	Any number of regions	CASAL2, GADGET, MULTIFAN-CL, SS3, WHAM
Stock-recruitment	Global and Local Density Dependence	CASAL2, GADGET
Fishery dynamics	Multiple fleets across regions and parameter sharing	CASAL2, GADGET, MULTIFAN-CL, SS3, WHAM, VPA-2Box
Demographics	Area-invariant or phenotypically-based	CASAL2, GADGET, SS3, WHAM, VPA-2Box
Data integration	Mark-recapture data	CASAL2, GADGET, MULTIFAN-CL, SS3, VPA-2Box
Random effects	Integration of random effects via Laplace Approximation	WHAM, SAM, FIMS
Simulation	Cohesive simulation testing (i.e., not an external package to the platform) via self-and cross-tests, and closed-loop functionality	WHAM, SAM, FIMS

<https://chengmatt.github.io/SPoRC/>



SPoRC Bridging From ADMB

- The bespoke ADMB was perfectly emulated in SPoRC (R-TMB)
 - Relative differences within $< 0.001\%$ for likelihood values, time-series estimates, selectivity, and key estimates (including reference points and management advice)
- Reviewed and approved by NPFMC Plan Team and Statistical and Scientific Committee
- Adopted for the basis of operational assessment in 2025



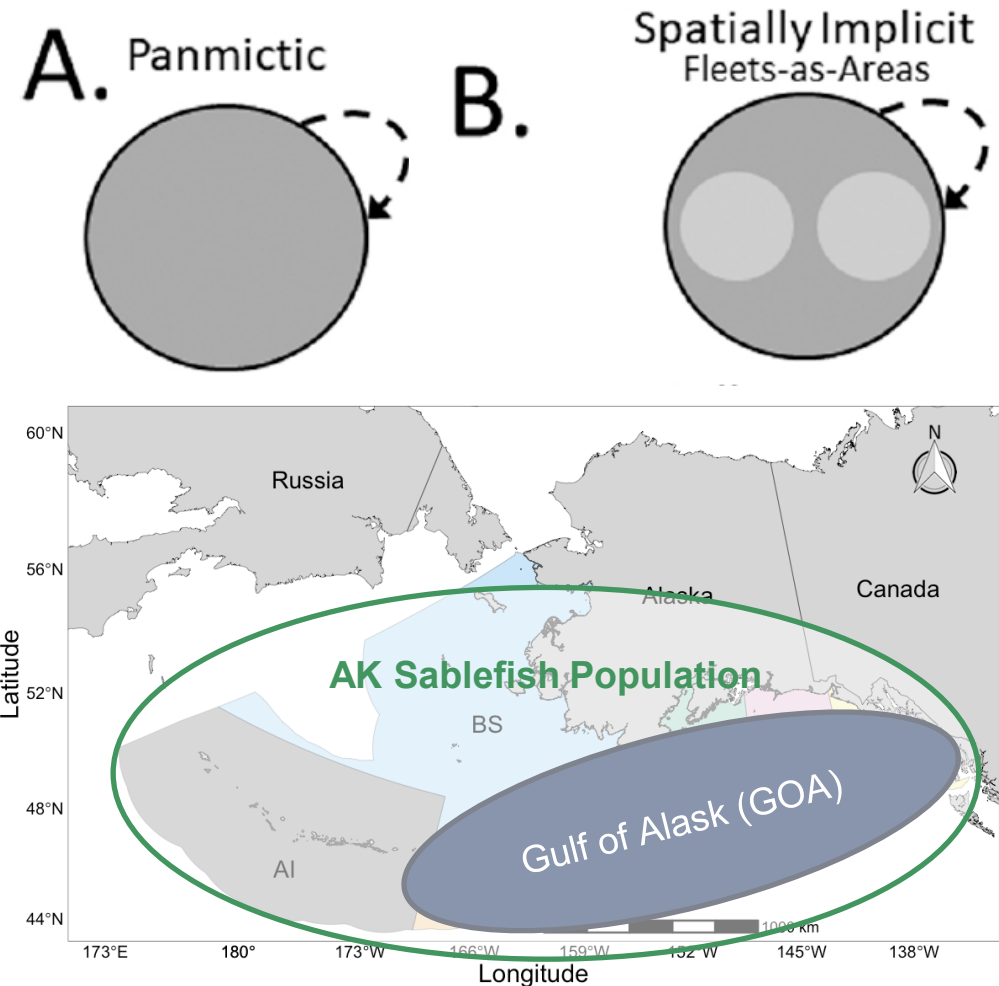
Data Integration and Observation Processes

	Data type	Statistical likelihood	Model treatment	Description
✓	Fishery removals	Lognormal	Fit with statistical likelihoods	Time series of removals (weight or numbers) by a fishery fleet
✓	Indices of abundance (or biomass; fishery and survey)	Lognormal	Fit with statistical likelihoods	Observed abundance or biomass by fishery or survey
✓	Age and/or length compositions (Fishery and survey)	Multinomial (available with Francis re-weighting), multinomial, and Logistic-normal	Fit with statistical likelihoods	Counts or proportions at age or length from fishery or survey
	Tagging	Poisson, Negative Binomial, Multinomial (Release and recapture-conditioned), Dirichlet-Multinomial (Release and recapture-conditioned)	Fit with statistical likelihoods	Counts of individuals (region, age, and sex) recaptured by both fishery and survey fleets
✓	Age-length transition matrix	—	Pre-specified input (assumed known)	Transition matrix to convert ages to lengths
	Survey environmental covariates	—	Pre-specified input (assumed known)	Environmental covariates used to model temporal changes in survey catchability
✓	Ageing Error	—	Pre-specified input (assumed known)	Ageing error used to fit to age-composition data
✓	Weight-at-age	—	Pre-specified input (assumed known)	Weight-at-age during the spawning season, fishery operations, and/or survey operations
✓	Maturity-at-age	—	Pre-specified input (assumed known)	Maturity-at-age to compute spawning stock biomass



Spatial and Fleet Structure

- Two primary models developed that differed only in fleet structure (implicit spatial structure):
 - 26.1_1Reg* – Single region panmictic model with no regional fleet structure
 - All data aggregated to a single region
 - Matches general structure of assessment currently used for management advice (25.12)
 - 26.2_FAA* – Single region Fleets-as-Areas (FAA) model that implicitly accounts for regional heterogeneity/availability/movement implicitly through NOAA Longline Survey fleet structure
 - All data aggregated to a single region **except** NOAA Longline Survey indices and age compositions
 - Disaggregated by region – BS, AI, and GOA
- Rest of presentation describes general framework as applies to both models



Data Summary



- New data for 2026 in **bold**
- Whale depredation by area held constant at 2022 levels
- AI and BS Regional NOAA Longline Surveys started in 1996 and 1997, respectively
- Francis reweighting applied (Method TA1.8; accounting for correlations among bins)

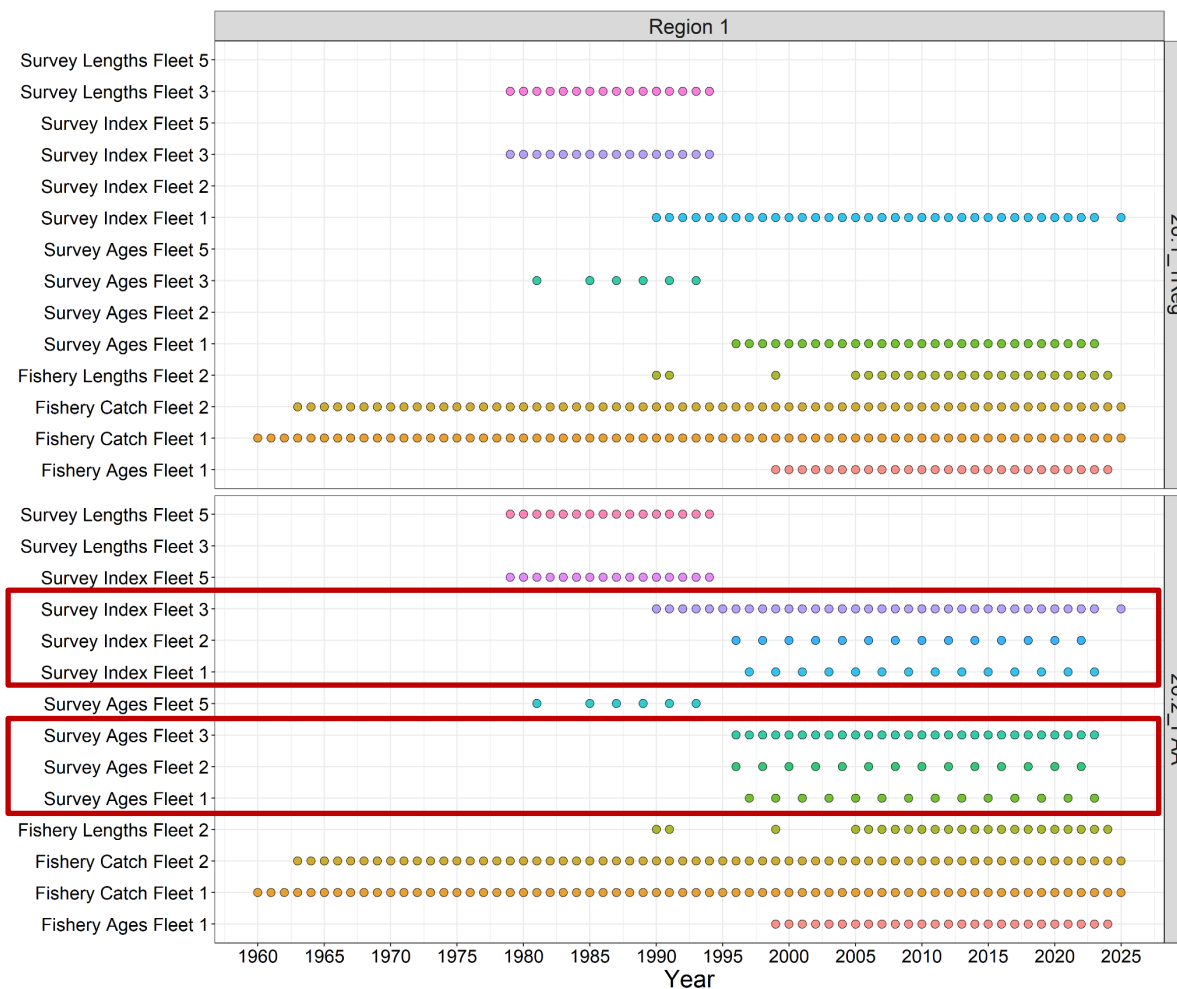
Source	Data	Years	Probability Distribution
Fixed Gear Fisheries	Catch	1960 – 2025	Lognormal
Trawl Fisheries	Catch	1960 – 2025	Lognormal
Non-Commercial Catch	Catch	1977 – 2025	Lognormal
U.S. Fixed Gear Fishery	Age	1999 – 2024	Multinomial (joint by sex)
U.S. Trawl Fisheries	Length	1990, 1991, 1999, 2005 – 2024	Multinomial (joint by sex)
Japanese Longline Survey	RPNs	1979 - 1994	Lognormal
	Length	1979 - 1994	Multinomial (joint by sex)
	Age	1981, 1985, 1987, 1989, 1991, 1993	Multinomial (joint by sex)
NOAA Longline Survey	RPNs	1990 – 2023, 2025	Lognormal
	Age	1996 – 2023	Multinomial (joint by sex)



Data Summary



Data



General Model Structure

- Single region (AK-wide)
- Sex-specific (males and females) dynamics:
 - Growth (size-age transition)
 - Selectivity
 - Natural Mortality (M)
 - 50:50 sex ratio
- Start year 1960
- Age structured (SCAA), projecting forward from initial ages and recruitment at age-2
- Assume yearly deviations from average recruitment
- Each fleet (fishery and survey) has independent, sex-specific selectivity (and regional LLS selectivity for the FAA model)
- Fishing mortality (F) estimated with yearly deviations for each fleet (fixed gear and trawl)
- Catchability (q) parameters freely estimated for each index (NOAA and Japanese LLS) and regionally for the FAA model

$$N_{r,y,a=2,s} = \mu^{Rec} \exp\left(\epsilon_{r,y}^{Rec} - \frac{\sigma_{Rec}^2}{2} b_y\right) \psi_s \zeta_r$$

$$N_{r,y+1,a+1,s} = N_{r,y,a,s} \exp(-Z_{r,y,a,s}), \quad \text{for } 2 \leq a < a_+$$

$$N_{r,y+1,a,s} = N_{r,y,a-1,s} \exp(-Z_{r,y,a-1,s}) + N_{r,y,a,s} \exp(-Z_{r,y,a,s}), \quad \text{for } a = a_+$$

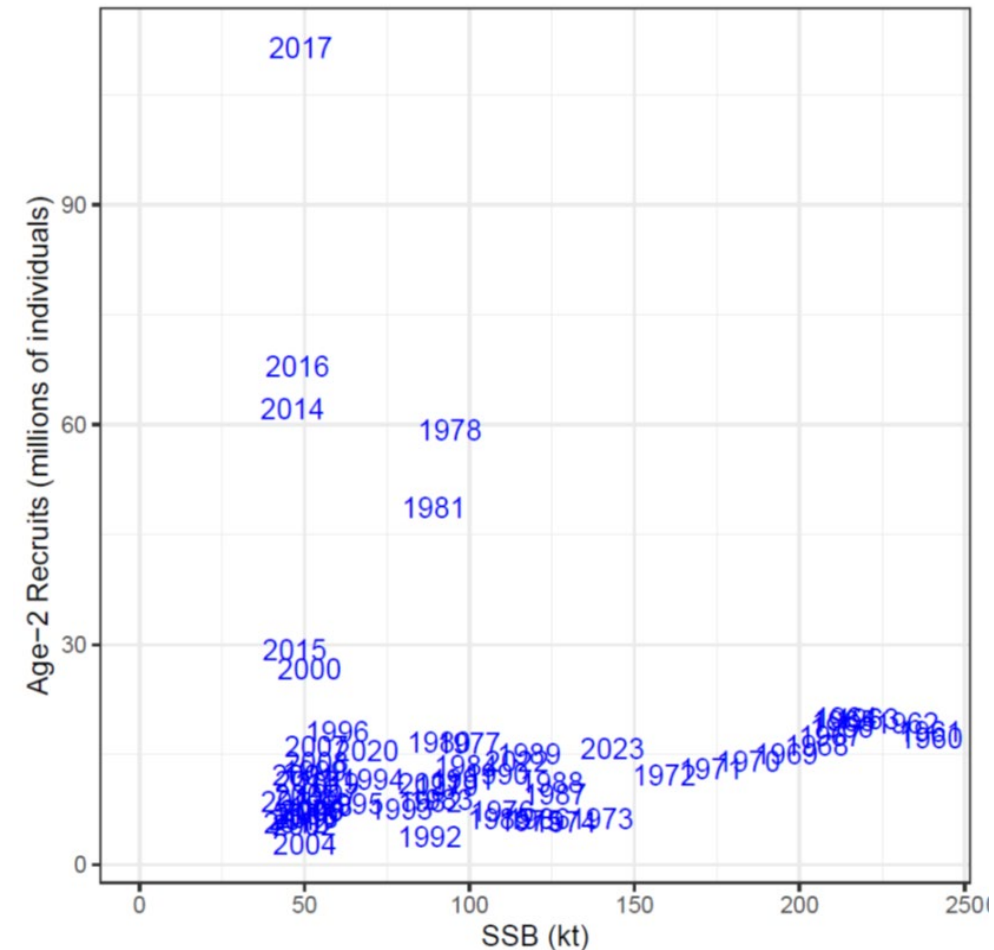
$$Z_{r,y,a,s} = Natmort_{r,y,a,s} + \sum_f Sel_{r,y,a,s,f}^{Fsh} Fmort_{r,y,f}$$

$$Fmort_{r,y,f} = \mu_{r,f}^{Fsh} \exp(\epsilon_{r,y,f}^{Fsh})$$



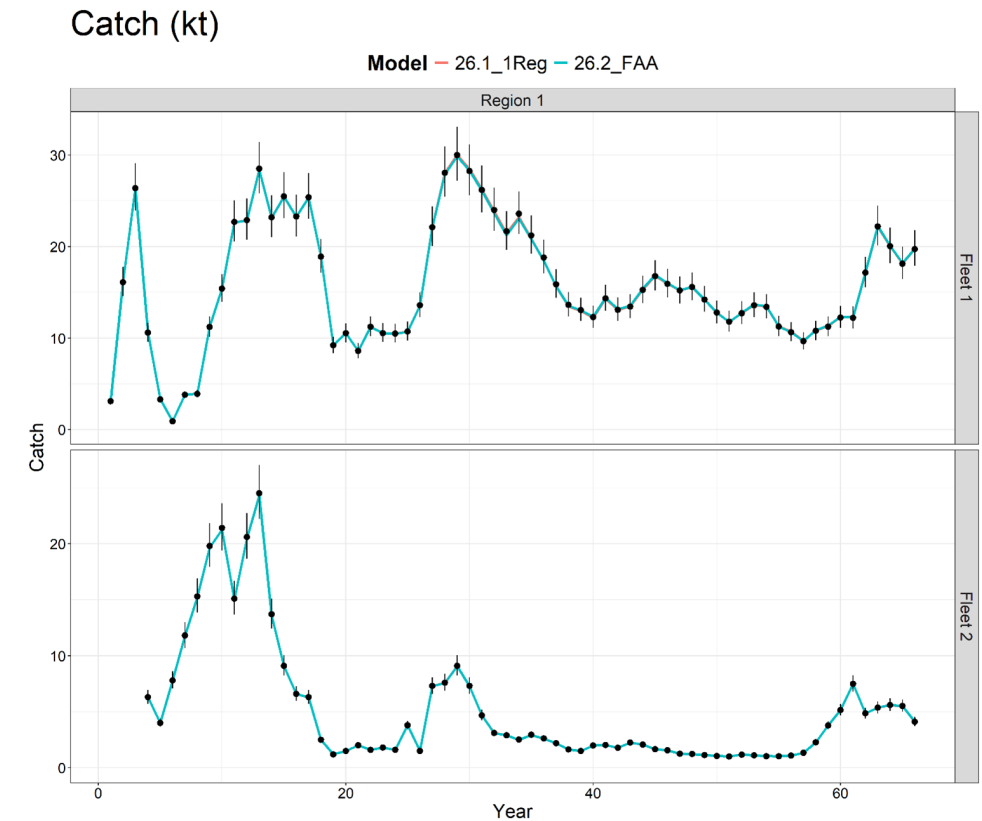
Biology, Recruitment, and Initial Age Structure

- Biological parameters input:
 - Size-age transition matrices (2 blocks pre- and post-1996)
 - Ageing error matrix
 - Weight-at-age
 - Maturity-at-age
- Natural mortality ‘estimated’ with a prior $\sim N(\ln(0.085), 0.1^2)$
- Equilibrium age structure with cohort deviations for each initial age
 - Initial exponential decay assumes 10% of mean fixed gear F
 - Age-specific deviations applied to equilibrium age structure
- Recruitment occurs at age-2 based on average recruitment with deviations
 - Lognormal bias correction integrated (Methot and Taylor, 2011)
 - No discernible stock-recruit relationship
 - High σ_R (0.9) to allow adequate estimation of spasmodic recruitment events



Fishing Dynamics

- Two fleets: fixed gear and trawl
- Fishing mortality estimated with yearly deviations for each fleet
- Logistic selectivity assumed for both fleets
 - Two parameters (i.e., age at 50% selectivity, $a_{50\%}$, and the slope of the curve, k) for each sex/fleet/time block
 - Estimated with loose priors $\sim N(\ln(4), 1.5^2)$
- Two time blocks assumed for the fixed gear fleet (single time block for trawl)
 - 1960 – 2015, 2016+
 - Corresponds to transition to pot gear (Cheng et al., 2025)
- Age compositions fit for fixed gear fleet and length compositions fit for the trawl fleet

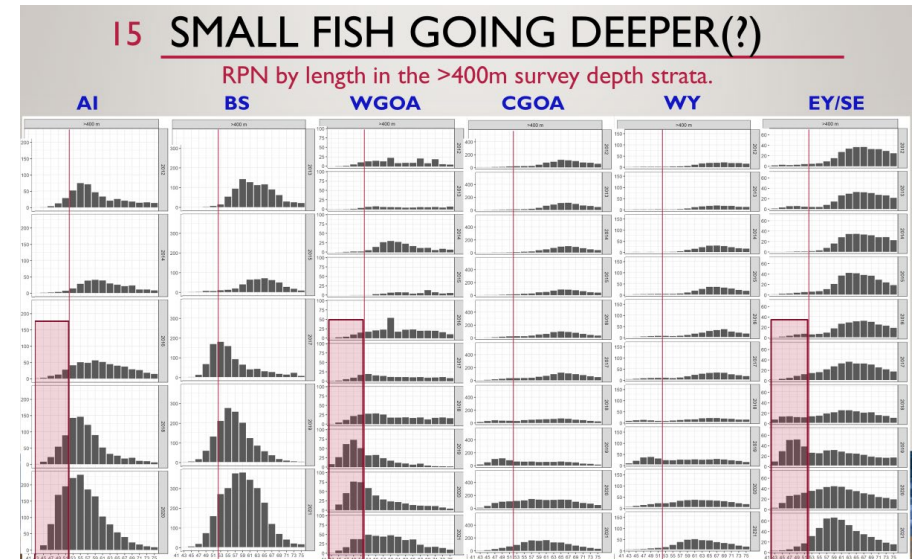
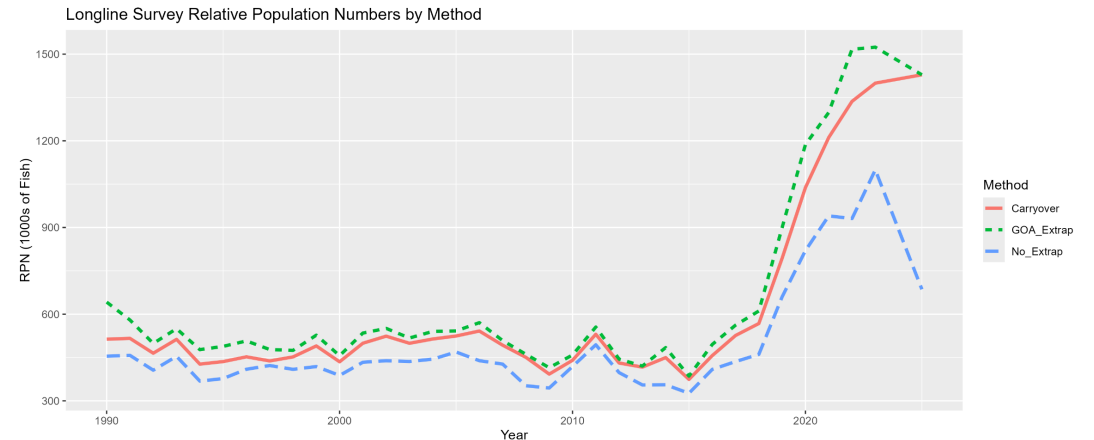


Confronting transitions in fishery fleet structure and selectivity: practical recommendations for integrated age-structured stock assessments based on simulation analysis

Matthew L.H. Cheng , Daniel R. Goethel , Peter-John F. Hulson , and Curry J. Cunningham 

Survey Dynamics – Single Region Model

- Two fleets: NOAA LLS and Japanese LLS
- Single catchability parameter estimated for each fleet
- Logistic selectivity assumed for both fleets
 - Estimated with loose priors $\sim N(\ln(3), 1.5^2)$
- Two time blocks assumed for the NOAA Longline Survey (single time block for Japanese LLS)
 - 1960 – 2015, 2016+
 - Corresponds to apparent changes in availability to LLS of small fish (small fish in deep strata where not historically observed)
- Age compositions fit for the NOAA LLS and both age and length compositions fit for the Japanese LLS

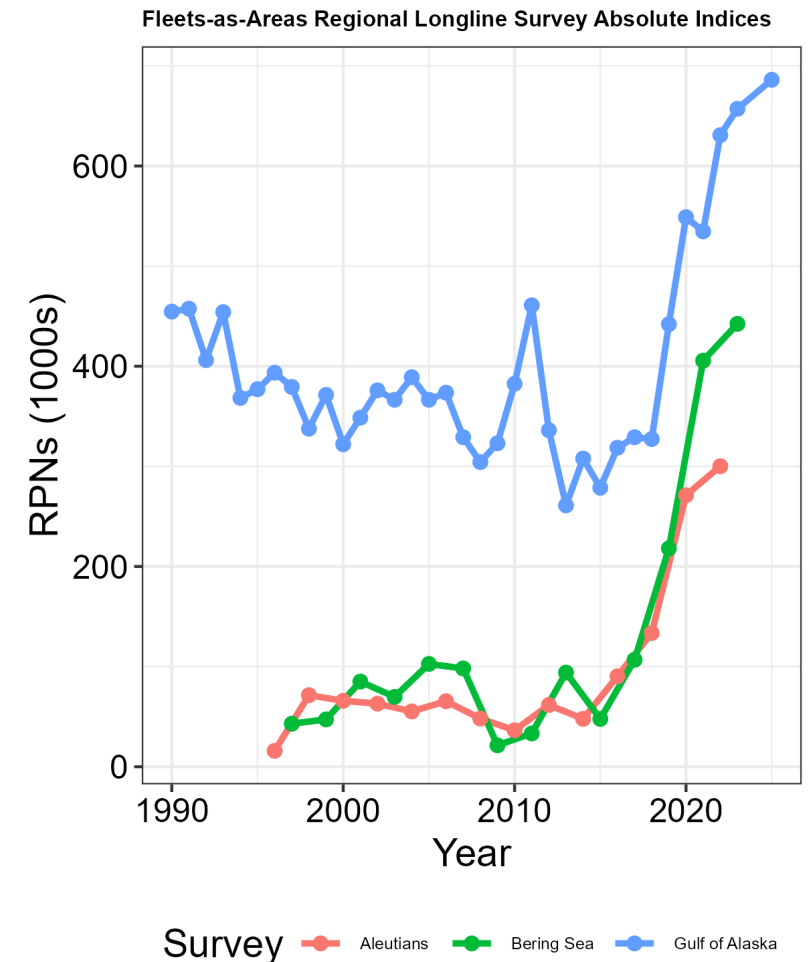


Heatwave
Impacts?



Survey Dynamics – FAA Model

- Structured the same as the single region model, but with 3 regional NOAA LLS fleets (BS, AI, GOA)
- Single catchability parameter estimated for each *regional* fleet
- Logistic selectivity assumed for all fleets
 - Estimated with loose priors $\sim N(\ln(3), 1.5^2)$
- Two time blocks assumed for each *regional* LLS fleet
 - 1960 – 2015, 2016+
- Age compositions fit disaggregated by *region* for each *regional* NOAA LLS fleet
- Total of 2 extra q parameters and 16 extra selectivity parameters estimated compared to single region model



Observation Models

- Catch assumes a fixed standard error of 0.05 (lognormal)
- Survey yearly standard errors calculated from the data (lognormal)
- Composition (age/length) Input Sample Sizes (ISS) scaled within fleets relative to observed yearly samples (multinomial)
 - Fleet-specific maximum values
 - Minimum ISS of 20
 - For FAA model, scaling of NOAA LLS ISS was across regions to maintain information on relative sample sizes across regions
- ESS determine from Francis reweighting

Data Source	26.1_IReg				26.2_FAA			
	SE	ISS	Data Wt	ESS	SE	ISS	Data Wt	ESS
Fixed Gear Catch	0.05	--	--	--	0.05	--	--	--
Trawl Catch	0.05	--	--	--	0.05	--	--	--
NOAA Longline Survey RPN BS	--	--	--	--	0.20	--	--	--
NOAA Longline Survey RPN AI	--	--	--	--	0.24	--	--	--
NOAA Longline Survey RPN GOA	--	--	--	--	0.08	--	--	--
NOAA Longline Survey RPN AK	0.07	--	--	--	--	--	--	--
Coop Survey RPN	0.07	--	--	--	0.07	--	--	--
Fixed Gear Age Composition	--	145	0.12	18	--	145	0.04	6
NOAA Longline Survey Age Composition BS	--	--	--	--	--	37	0.51	19
NOAA Longline Survey Age Composition AI	--	--	--	--	--	39	1.91	75
NOAA Longline Survey Age Composition GOA	--	--	--	--	--	77	0.29	23
NOAA Longline Survey Age Composition AK	--	171	0.28	47	--	--	--	--
Japanese Longline Survey Age Composition	--	83	0.17	14	--	83	0.18	15
Japanese Longline Survey Size Composition	--	55	0.34	19	--	55	0.33	18
Trawl Fishery Size Composition	--	45	0.25	12	--	45	0.14	7



Priors and Estimated Parameters

- MLE with penalized likelihood used to estimate parameters
- Priors:
 - Recruitment deviations $\sim N(0, \sigma_R^2)$
 - Fishing mortality deviations $\sim N(0, 0.75^2)$
 - Natural mortality $\sim N(\ln(0.085), 0.1^2)$
 - Fishery selectivity parameters $\sim N(\ln(4), 1.5^2)$
 - Survey selectivity parameters $\sim N(\ln(3), 1.5^2)$
- Total estimated parameters
 - Single region: 253
 - FAA: 271

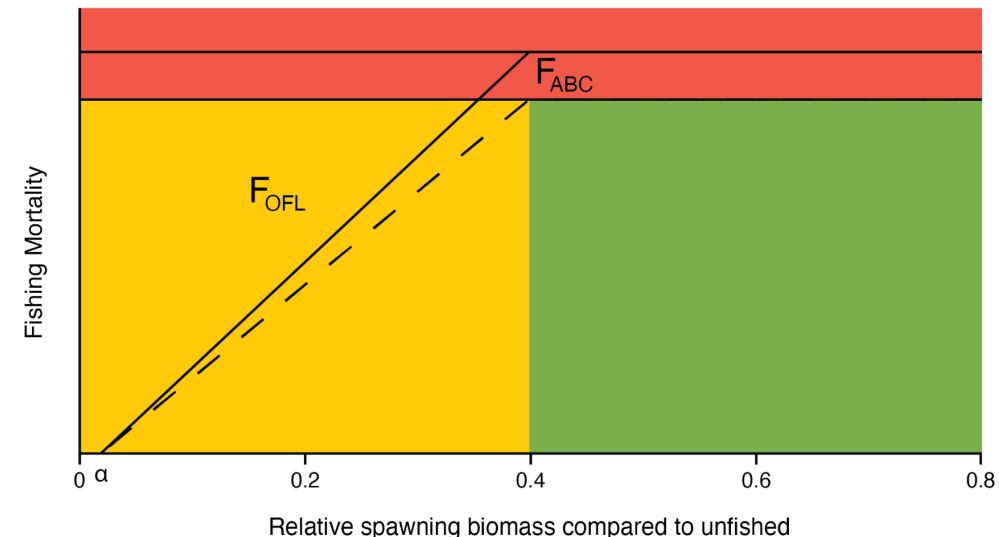
Parameter Name	25.12_1Reg_Cont	26.1_1Reg	26.2_FAA
Catchability	5	2	4
Mean recruitment	1	1	1
Natural Mortality	1	2	2
Initial Deviations	28	28	28
Recruitment Deviations	65	65	65
Mean Fishing Mortality	2	2	2
Fishing Mortality Deviations	129	129	129
Fishery Selectivity	14	12	12
Survey Selectivity	11	12	28
Total	256	253	271



Catch Projections and Reference Points

- Standard methods for reference points (per-recruit) and associated catch projections used by all AFSC assessments
- Parameters (e.g., relative fishing mortality among fishing fleets, selectivity, natural mortality, and biological inputs) are taken from the terminal year of the assessment
- Biomass reference points (B40%) are calculated from per-recruit reference points based on average recruitment (1979 – terminal assessment year - 2)
 - F_{ABC} is defined as F40% and F_{OFL} is defined as F35%, except when reduced by the NPFMC threshold/sloped HCR due to SSB falling below SSB_{OFL} (B35%)
- For projections:
 - Start from the terminal year numbers-at-age
 - Recruitment is simulated as an inverse Gaussian distribution parameterized using based on average recruitment and variance (1,000 simulations)
- A standard set of seven projections is required for each stock

Groundfish Control Rule



Summary

- All models developed in the SPoRC generalized assessment package
- Two primary models to be compared:
 - 26.1_1Reg: Single region panmictic model
 - 26.2_FAA: FAA model with regional NOAA LLS fleets
- Structure between the two models are essentially identical except for LLS fleet parametrization (and associated data fitting)
- Rationale for the two models and associated model building steps (including simulation analyses) to be presented tomorrow

